

FSO COMMUNICATION SYSTEM

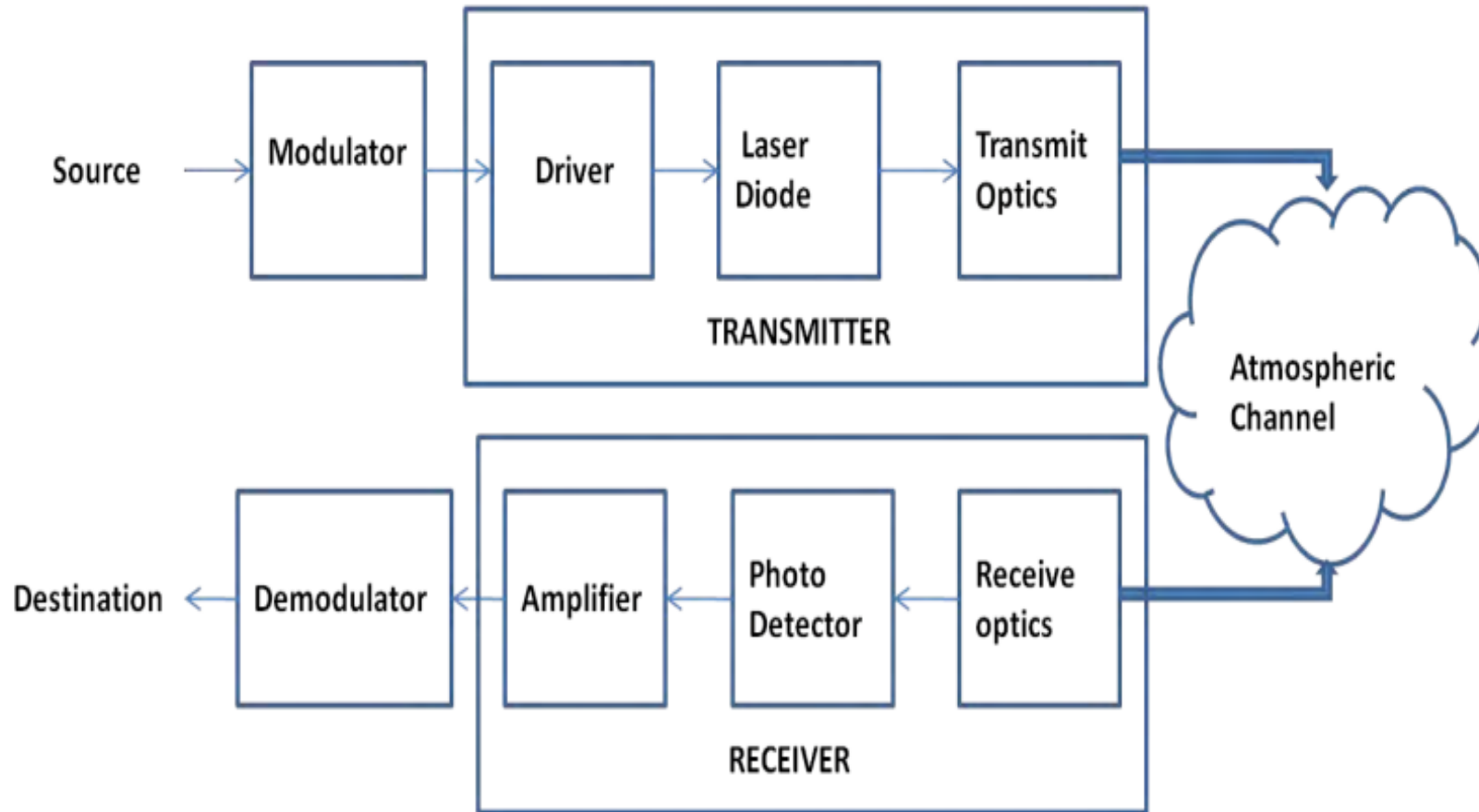
ABOUT FSO

- A wireless technology that uses light beams from lasers or LEDs to transmit data through the air, outer space or a vacuum.
- Requires a straight, clear path between the sender and receiver.
- Heavy fog, rain, snow or dust can scatter the light beam and weaken the signal

COMMON USES:

- Space and Satellites
- Urban Networking
- Temporary or Emergency Links

BLOCK DIAGRAM



SOURCE

- Produces the original information.
- Information may be text, voice, image or video.
- Sends the data for processing.

CHANNEL ENCODER

- Adds redundant bits using mathematical calculations.
- Improves communication reliability.
- Helps detect and correct transmission errors.
- Protects data against atmospheric impairments.

Examples: Low Density Parity Check (LDPC), Reed–Solomon, Turbo, Polar.

MODULATOR

- Receives encoded bits.
- Maps bits into optical pulses.
- Makes the data suitable for laser transmission.
- Common methods: On-Off Key, Pulse Position Modulation, Orthogonal Frequency-Division Multiplexing.

DRIVER

- Provides the required driving current.
- Converts modulation signals into laser drive signals.
- Ensures stable laser operation.
- Protects the laser from excessive current.

LASER DIODE

- Converts electrical signals into laser light.
- Produces a narrow optical beam.
- Supports high-speed communication.
- Common wavelengths: 850 nm and 1550 nm.

TRANSMIT OPTICS

- Focuses the laser beam.
- Reduces beam divergence.
- Improves transmission efficiency.
- Directs the beam toward the receiver.

ATMOSPHERIC CHANNEL

- Laser travels through free space.
- Introduces attenuation.
- Causes scattering and absorption.
- Produces atmospheric turbulence.
- Affected by fog, rain, dust and smoke.
- May introduce pointing errors.

RECEIVE OPTICS

- Collects the transmitted optical signal.
- Focuses light onto the photodetector.
- Increases received optical power.
- Improves signal collection.

PHOTO DETECTOR

- Receives the laser beam.
- Converts light into electrical current.
- Produces the received electrical signal.
- Common devices: Photodiodes

AMPLIFIER

- Amplifies weak signals.
- Improves signal level.
- Increases Signal-to-Noise Ratio (SNR).
- Prepares the signal for processing.

CHANNEL ESTIMATION

- Uses known pilot symbols.
- Estimates channel gain.
- Measures attenuation.
- Estimates noise level.
- Evaluates turbulence effects.
- Does not recover the transmitted data.

EQUALIZER

- Uses channel estimation results.
- Compensates channel attenuation.
- Reduces distortion.
- Improves signal quality.
- Produces a cleaner signal.

DEMODULATOR

- Detects transmitted symbols.
- Converts optical symbols into bits.
- Makes bit decisions.
- Produces the received bit stream.

CHANNEL DECODER

- Uses redundant bits added by the encoder.
- Detects bit errors.
- Corrects transmission errors.
- Recovers the original information bits.
- Improves communication reliability.

DESTINATION

- Displays the received data.
- Final user receives the information.
- Communication process ends.

THANK YOU